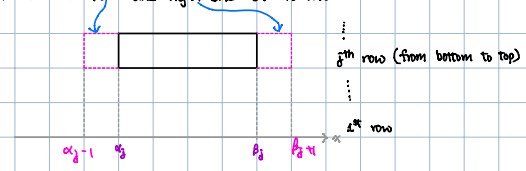
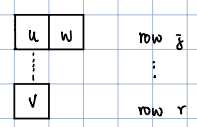
Def: For $\alpha, \beta \in \mathbb{Z}^{\ell}$ s.t. $\alpha_j \le \beta_j$ $\forall\, 1 \le j \le \ell$, define

β/α = tuple of single row skew shapes β_j/α_j s.t. the left end has x-coordinate α_j and the right end has x-coordinate β_j.

"virtual" boxes adjacent to the left and right end of the row :

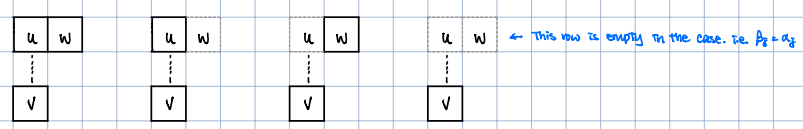


Def: Given a tuple of skew row shapes β/α, three boxes (u, v, w) form a ω_0-triple when



(u, w can be "virtual" boxes)

Hence possible ω_0-triples are



Notation: $h_{\omega_0}(\beta/\alpha)$ = # ω_0-triples in β/α (To count, we fix a "real" box and count the # left virtual or real boxes above it.)

i.e. fix v

There are choices of u

These boxes must contain a box (may be virtual) on its right i.e. w exists

choices of v

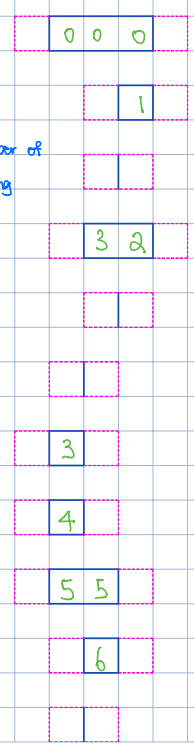
By definition, $h_{\omega_0}(\beta/\alpha) = \sum_{1 \le r < j \le \ell} |[\alpha_r+1, \beta_r] \cap [\alpha_j, \beta_j]|$

choices of u

e.g. $\beta = (1, 2, 2, 1, 1, 1, 2, 3, 2, 3, 3)$

$\alpha = (1, 1, 0, 0, 0, 1, 2, 1, 2, 2, 0)$

(number in each real box (v) counts the number of left virtual / real box above it, i.e. counting choices of u to form a triple with v)



$\therefore\ h_{\omega_0}(\beta/\alpha)$ = sum of the numbers

$= 29$

Def: For a totally ordered alphabet $\mathcal{A}$, a row strict tableau of shape β/α is a map

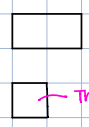
$$S : \beta/\alpha \to \mathcal{A} \quad \text{s.t. each row is strictly increasing.}$$

Convention: $\begin{cases} S(x) = -\infty & \text{if } x \text{ is a left virtual box} \\ S(x) = +\infty & \text{if } x \text{ is a right virtual box} \end{cases}$

Notation: $RST(\beta/\alpha, \mathcal{A})$ = set of all row strict tableau of shape β/α in alphabet $\mathcal{A}$

Set $RST(\beta/\alpha, \mathcal{A}) = \phi$ if $\alpha_j > \beta_j$ for some $1 \le j \le \ell$.

Def: A ω_0-tuple (u,v,w) is an increasing ω_0-triple in S if $S(u) < S(v) < S(w)$. $h_{\omega_0}(S)$ = # increasing ω_0-triples in S.



This value is in the "interval" formed by $(S(u), S(v))$

e.g.

$3 \in (2,5)$

Increasing ω_0-triple

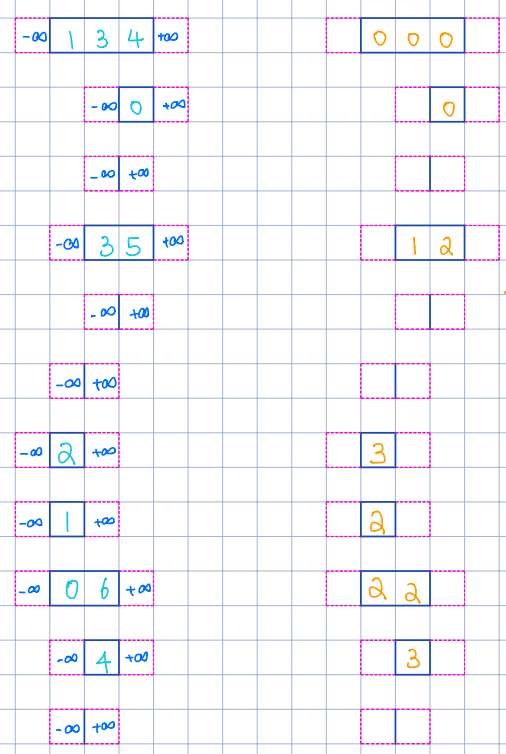
$2 \notin (2,5)$

NOT an increasing ω_0-triple

e.g. $\beta = (1,2,2,1,1,1,2,3,2,3,3)$

$\alpha = (1,1,0,0,0,1,2,1,2,2,0)$

$S =$



← The number in each real box (v) counts the number of 'u' forming an increasing ω_0-triple with the fixed v box.

$\therefore h_{\omega_0}(S)$ = sum of these numbers $= 15$.

Def: For $S \in RST(\beta/\alpha, \mathbb{N})$, define

$$x^{wt_+(S)} := \prod_{\substack{b \in \beta/\alpha \\ S(b) \neq 0}} x_{S(b)} \quad , \quad x^{wt(S)} := \prod_{b \in \beta/\alpha} x_{S(b)} .$$

$$x^{wt(S)} = x_0^{\#0 \text{ in } S} \cdot x^{wt_+(S)}$$

e.g. In the example above,

$$x^{wt_+(S)} = x_1^2 x_2 x_3^2 x_4^2 x_5 x_6 \quad , \quad x^{wt(S)} = x_0^2 x_1^2 x_2 x_3^2 x_4^2 x_5 x_6 = x_0^2 \, x^{wt_+(S)}$$

Def : For $\alpha, \beta \in \mathbb{N}^m$, define

$$N_{\beta/\alpha} = N_{\beta/\alpha}(X;q) := \sum_{S \in RST(\beta/\alpha, \mathbb{Z}^+)} q^{h_\omega(S)} x^{wt(S)}$$

In this case $wt(S) = wt_+(S)$

* In Prop 4.5.2 (Path), $N_{\beta/\alpha} \in \Lambda$ with Schur expansion involving only s_λ with $\ell(\lambda) \leq m$.